

Mathieu Néron

SOFTWARE ENGINEER

San Francisco, CA

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Summary

Software engineer with 15+ years shipping at scale. Currently building fraud-risk infrastructure at Highnote, processing **3M+ monthly card transactions** (~\$350M+ in monthly settlement volume) across **~45 enterprise programs** and **1.2M+ cardholders**; previously Senior → LMTS at Tableau, building authoring and server features used by **87K+ customer companies** and **3M+ Tableau Public creators**. Background spans fintech, data visualization, and e-signature; building applied-ML fluency hands-on (**Medea**: multi-modal PyTorch classifier with RAG explanation, F1 0.90 on channel-grouped CV).

Experience

Highnote

San Francisco, CA

SENIOR SOFTWARE ENGINEER

Aug. 2023 – Present

- *Java, TypeScript, GraphQL, GCP, Spanner, BigQuery, Temporal, Elasticsearch, Python, ANTLR*
- Owned **OFAC sanctions screening** on Highnote's synchronous money-movement critical path (Visa Direct / RTP / ACH); landed a three-layer optimization — in-memory early-exit, False/True positive lookups deferred to candidate matches, Spanner bounded-staleness reads — to hit a **P99 ≤ 100 ms** SLA powering **Ferry's launch as Highnote's first Instant Payments subscriber** (Dec 2025); **99.996%** of ~244K monthly screenings now bypass the database entirely.
- Designed and shipped a customizable **geofencing rule** now deployed across **7 subscribers** (19 rule instances) on Highnote's rule engine; mid-2025 tuning lifted true-positive rate **3–4×** on the Fleet-program enforcement path.
- Built the **real-time velocity feature store** powering Highnote's fraud rule engine — intra-hour merchant/BIN/card auth-velocity counters (count + amount) written on the synchronous auth path and served to rule evaluation via a CQRS read API; collapsed the Redis read from N round-trips to **2 concurrent MGETs (~1 RTT)**, engineered to hold **P95 < 100 ms under load** as the program scales.
- Led migration of the core SMS stack from AWS Pinpoint to **Twilio** and built the platform's first **bi-directional SMS fraud-detection workflow** on **Temporal** — conversation state lives in Temporal's durable workflows, so the SMS layer uses Twilio's lightweight Messaging API rather than the heavier and pricier Conversations API; anchors a flagship enterprise rollout.
- Drove adoption of **Temporal** (durable workflows) and **Elasticsearch** as first production use cases at Highnote; both now standard across multiple teams.
- Extended the **ANTLR grammar** powering Highnote's natural-language query layer with fuzzy / typo-tolerant search.

Tableau (Salesforce)

Palo Alto, CA

SENIOR SOFTWARE ENGINEER → LMTS

Jun. 2019 – Jul. 2023

- *React, Redux, TypeScript, Java, C++, Qt* — Promoted to Lead Member of Technical Staff in 2020.
- Onboarded a new AI-predictive-model content type in Tableau Server for **Salesforce Einstein Discovery** integration — contributing to the server stack delivering **80B+ AI-powered predictions per day**.
- Drove the **Object Graph** view in Tableau's Data pane from beta to GA — the default relationships editor in Tableau's multi-table data model, part of the authoring stack behind Tableau Public's **7M+ vizzes** from **3M+ creators**; shipped transition animations, tooltips, sorting, and re-rooting.
- Contributed to the **Data Preview Area** hybridization — metadata grid plus embedded relationship editor in a single surface.

Paysafe

Montréal, Canada

JAVA DEVELOPER

2016 – 2019

- *Angular 2+, TypeScript, Java, Spring, Oracle*
- Shipped the new Angular 2+ **Merchant Portal** frontend from greenfield.
- **Cut frontend test runtime 6×** by migrating from PhantomJS to Nightmare.

eSignLive

Montréal, Canada

JAVA DEVELOPER

2014 – 2016

- *Java, Spring, Guava, AWS, MySQL, Jenkins*
- **Cut Windows document converter downtime by 50%**.
- Integrated Equifax, Twilio, PDFBox, and Google Noto Fonts.

Avanade

Montréal, Canada

APPLICATION DEVELOPER

2012 – 2014

- Enterprise web engineering for Canadian National Railway (*www.cn.ca* redesign on Sitecore / ASP.NET) and Hydro-Québec (internal time-tracking app; led frontend-stack selection — SAPUI5 + Infragistics).

Projects

github.com/mathieu-neron/RealTube

Personal learning project

REALTUBE — CROWDSOURCED AI-GENERATED YOUTUBE CONTENT FILTER

2026

- Full-stack browser extension (Chrome/Firefox, Manifest V3) for crowdsourced flagging and hiding of AI-generated YouTube content across feed, search, recommendations, and Shorts.
- Built as a hands-on Go + Python learning exercise: **Go** (Fiber v3) on the low-latency lookup path, **Python + FastAPI** for the inference-adjacent services, **TypeScript + React 19** extension, PostgreSQL + Redis + Docker + Prometheus.
- Privacy-engineered URL lookup using a **k-anonymity hash-prefix** scheme (HIBP-style) so the backend never sees full URLs in plaintext; contract-tested API and end-to-end test coverage.

github.com/mathieu-neron/medea

Personal learning project (natural follow-up to RealTube)

MEDEA — MULTI-MODAL AI-GENERATED YOUTUBE CLASSIFIER

2026

- Built an end-to-end ML pipeline to learn the full applied-ML stack hands-on: yt-dlp ingest → four-modality feature extraction (**CLIP ViT-B/32** visual, **faster-whisper + wav2vec2** anti-spoofing audio, **MiniLM** text, handcrafted scalars) → **Chroma** vector DB → small **PyTorch** MLP head → FastAPI server with retrieval-augmented **local-LLM rationale** via Ollama.
- Used **channel-grouped LeaveOneGroupOut CV** (10 folds, ~94-clip dataset) after UMAP showed channels as 10 distinct clusters — random video-level splits would have memorized channel identity, not “AI-ness.” Final MLP **F1 0.90 / ROC-AUC 0.96** on this honest split.
- Diagnosed a silent-audio failure mode (anti-spoof model misfires on no-speech clips) and fixed it with `has_speech × ai_voice_prob` interaction features (F1 0.87 → 0.90); added a **temporal prior cap** for pre-2017 uploads to bound MLP extrapolation outside the training cone.

Skills

Languages	Java, TypeScript/JavaScript, Python, Go, C/C++, C#
Frameworks	Spring, Temporal, Node.js, React/Redux, Angular, GraphQL, REST, Qt, .NET, ANTLR
Platforms	GCP, AWS, Spanner, BigQuery, PostgreSQL, Elasticsearch, Redis, Oracle, MySQL, Docker
ML / AI	PyTorch, Hugging Face Transformers, sentence-transformers, CLIP, Whisper, Chroma, MLflow, Ollama, RAG
Spoken	French (native), English (fluent)

Education

Université de Sherbrooke

Sherbrooke, Canada

B.ENG., COMPUTER ENGINEERING

2007 – 2011

- Specialization in Software Engineering and Artificial Intelligence.

Internships

Electronic Arts Canada

Burnaby, Canada

SOFTWARE ENGINEER INTERN

Summer 2011

- Built a distributed C++ / C# load-test harness running thousands of concurrent bots against **SSX**'s multiplayer server; used internally and demoed at **Gamescom 2011**.

Autodesk | Ubisoft | Lockheed Martin

Canada

EARLIER INTERNSHIPS

2008 – 2010

- **Autodesk** (2010) — animation plug-in inside Maya (C++ / Qt). **Ubisoft** (2009) — QA automation (C# / .NET). **Lockheed Martin** (2008) — software evaluation.